A Matched inputs

Sample reads

AID-null control reads

Matched preparation and run
Depth-matched usable reads

B Control-informed calibration

Observed change versus control-derived background

Candidate absent from controls

Evidence can exceed background

Candidate recurrent in controls

Evidence can match background

Candidate tests → empirical decoys

Controls set the calling threshold
Empirical false-discovery target: 5%

C Interpretable calls

Retained variants

Mutation burden
Substitution spectrum
Position and hotspots
Clones and reconstruction

D Platform-specific measurement routes

Nanopore / Illumina

Control-calibrated variants
Optional UMI consensus

Sanger clones

Count changes at Q ≥ 20
Report background alongside

PacBio switch reads

Junction structure
Microhomology and insertions

Schematic pileups illustrate the decision logic; colours distinguish candidate changes and recurrent background.